EC-360 Software Engineering

Sprint-1 (P-2) – Controlled Agile Execution

(Group Assignment)

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1. Purpose

Sprint-1 evaluates your team's ability to **apply Scrum in practice**, not to complete the full system.

The focus is on:

- Proper backlog engineering (FR + NFR + AC)
- Structured Sprint execution
- Git & Jira discipline
- Working software increment
- Reflection and improvement

Sprint Duration: **2 Weeks (2 Mar – 13 Mar) – you can start earlier as well. i.e., (23 Feb – 6 Mar)**

Viva/Demo: On date Specified on LMS

2. What Sprint-1 Must Demonstrate

Sprint-1 must clearly show application of concepts from:

Week-2 (Agile & Scrum)

- Sprint Goal defined
- Stories estimated with Story Points
- Sprint Backlog created
- Daily Scrum conducted
- Sprint Review & Retrospective completed
- Basic velocity reflection (Committed vs Completed)

Week-3 (Configuration Management & Git)

- GitHub repository used properly
- Feature branches used
- Pull Requests raised before merge
- Commits linked with Jira issue keys
- No direct commits to main branch

Week-4 (Requirements Engineering)

- User stories derived from stakeholder needs
- Each selected story includes:
  - Functional Requirement (FR)
  - At least one measurable Non-Functional Requirement (NFR)
  - Acceptance Criteria (testable)
- Stories meet Definition of Ready (DoR) before entering sprint

- Definition of Done (DoD) followed before closing story

3. Sprint-1 Lifecycle Expectations

- **Before Sprint**
  - Product Backlog refined & prioritized
  - Stories meet DoR
  - Roles clearly assigned
  - Jira & GitHub configured
- **Sprint Planning**
  - Define Sprint Goal (one sentence)
  - Select realistic stories
  - Estimate using story points
  - Create Sprint Backlog in Jira
- **During Sprint**
  - Conduct Daily Scrum
  - Update Jira board regularly
  - Commit code frequently
  - Raise PRs before merging
  - Maintain traceability
- **Sprint Review**
  - Demonstrate working software (live)
  - Show Jira board progress
  - Show GitHub commit history
- **Sprint Retrospective**
  - Identify what went well
  - Identify what went wrong
  - Define at least 2 improvement actions

4. Minimum Engineering Expectations

Sprint-1 increment must include:

- Working functional feature (not only UI)
- At least one implemented and validated NFR
- Use of backend logic or validation mechanism
- Clean integration between components
- Code merged properly through PR

5. Definition of Done (Sprint-1)

A story is Done only if:

- Acceptance Criteria satisfied

- Code reviewed and merged via PR
- NFR considered
- No open conflicts
- Jira updated

6. Sprint-1 Evaluation Rubric (15 Marks)

The rubric below is outcome-based and broad to allow efficient grading.

A. Requirements Engineering Quality (3 Marks)

Level	Description
3	Stories well-structured, clear AC, measurable NFR included, DoR applied
2	Stories mostly clear, minor gaps in AC or NFR clarity
1	Weakly written stories, vague AC, unclear NFR
0	No structured backlog evidence

B. Scrum Execution & Sprint Management (4 Marks)

Level	Description
4	Clear Sprint Goal, estimation applied, board updated, velocity reflected, retrospective documented
3	Most Scrum practices followed with minor gaps
2	Partial Scrum adoption, limited reflection
1–0	No clear Scrum structure

C. Git & Jira Discipline (4 Marks)

Level	Description
4	Proper branching, PR usage, commits linked to Jira, clear traceability
3	Mostly correct Git usage, minor discipline gaps
2	Inconsistent branching or weak traceability
1–0	Direct commits to main, no PRs, no linkage

D. Working Increment & Engineering Depth (4 Marks)

Level	Description
4	Working increment delivered, AC demonstrated, NFR validated, meaningful implementation depth
3	Mostly functional increment with minor issues
2	Limited functionality or weak NFR implementation
1–0	No working increment or superficial demo

7. Final Grade Mapping

Marks	Interpretation
13–15	Strong Agile & Engineering Execution
10–12	Good Sprint Implementation
7–9	Partial Scrum Adoption
≤6	Weak Process & Technical Discipline

8. Important Notes

- Roles must rotate in next sprint
- Git commits must reflect individual contribution
- Jira evidence is mandatory
- Sprint-1 artifacts will be reused in Final Report
- Real client project may earn +1 absolute mark

9. One-Line Reminder for Students

Your grade depends on how well you apply Scrum, engineer quality backlog items, maintain Git discipline, and deliver a working increment — not on feature count.

---------- BEST OF LUCK ----------